1. **Write a Python program to generate a complete summary of an employee dataset showing:**

- **Number of rows**
- **Number of columns**
- **Data types**
- **Missing values**
- **Statistical summary**

Dataset: employee_data.csv

```
Employee_ID,Name,Age,Department,Salary
E1,Alice,25,HR,45000
E2,Bob,30,IT,55000
E3,Charlie,28,Finance,
E4,David,,IT,60000
E5,Eva,35,HR,52000
```

Code

```python
import pandas as pd
df = pd.read_csv("employee_data.csv")
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("\nData Types")
print(df.dtypes)
print("\nMissing Values")
print(df.isnull().sum())
print("\nStatistical Summary")
print(df.describe(include="all"))
```

Output

```
PS D:\FDS>  & 'C:\Program Files\Python312\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '5187
0' '--' 'D:\FDS\Hands-On_DS1.py'
Number of Rows: 5
Number of Columns: 5

Data Types
Employee_ID        str
Name               str
Age            float64
Department         str
Salary         float64
dtype: object

Missing Values
Employee_ID    0
Name           0
Age            1
Department     0
Salary         1
dtype: int64

Statistical Summary
       Employee_ID   Name        Age Department        Salary
count            5      5   4.000000          5      4.000000
unique           5      5        NaN          3           NaN
top             E1  Alice        NaN         HR           NaN
freq             1      1        NaN          2           NaN
mean           NaN    NaN  29.500000        NaN  53000.000000
std            NaN    NaN   4.203173        NaN   6271.629241
min            NaN    NaN  25.000000        NaN  45000.000000
25%            NaN    NaN  27.250000        NaN  50250.000000
50%            NaN    NaN  29.000000        NaN  53500.000000
75%            NaN    NaN  31.250000        NaN  56250.000000
max            NaN    NaN  35.000000        NaN  60000.000000
PS D:\FDS>
```

2. **Write a Python program to calculate the frequency distribution of students' grades and display the results using a bar chart.**

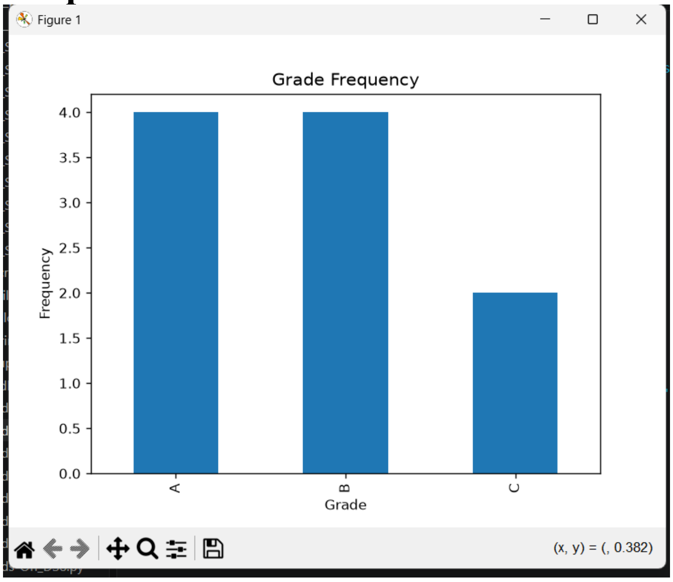
Dataset: student_grades.csv

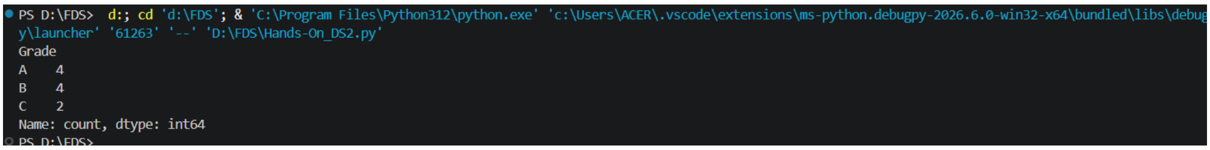
```
Student_ID,Grade
S1,A
S2,B
S3,A
S4,C
S5,B
S6,A
S7,B
S8,C
S9,A
S10,B
```

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student_grades.csv")

frequency = df["Grade"].value_counts().sort_index()

print(frequency)

frequency.plot(kind="bar")

plt.title("Grade Frequency")
plt.xlabel("Grade")
plt.ylabel("Frequency")
plt.show()
```

Output





3. **Write a Python program to plot the histogram of house prices and identify whether the data is normally distributed or skewed.**

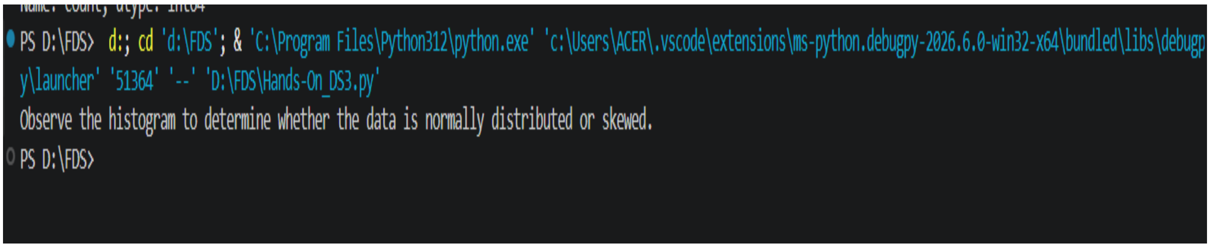
Dataset: house_prices.csv

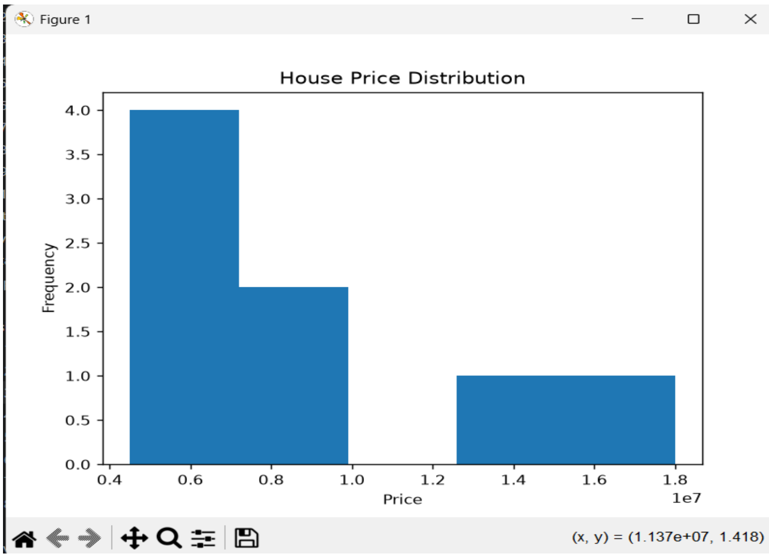
House_ID,Price
H1,4500000
H2,5200000
H3,6100000
H4,7000000
H5,7500000
H6,8000000
H7,15000000
H8,18000000

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("house_prices.csv")

plt.hist(df["Price"], bins=5)

plt.title("House Price Distribution")
plt.xlabel("Price")
plt.ylabel("Frequency")

plt.show()

print("Observe the histogram to determine whether the data is normally distributed or skewed.")
```

Output





4. **Write a Python program to create a box plot of salary data and identify the presence of outliers.**

Dataset: salary_data.csv

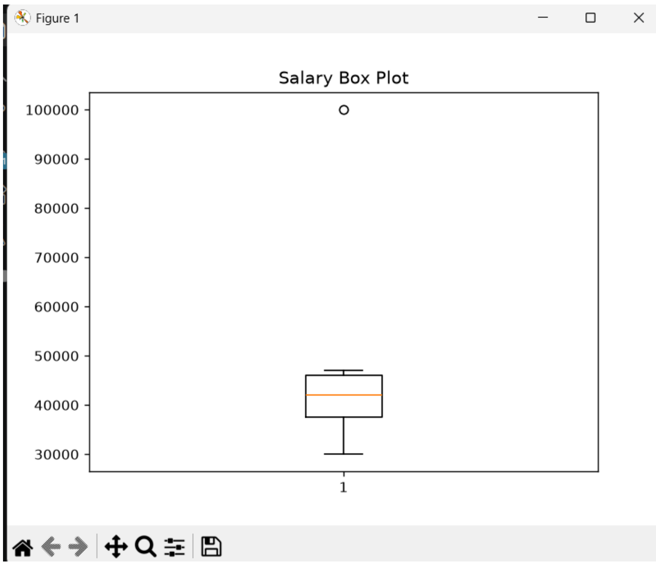
Employee_ID,Salary
E1,30000
E2,35000
E3,40000
E4,42000
E5,45000
E6,47000
E7,100000

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("salary_data.csv")

plt.boxplot(df["Salary"])

plt.title("Salary Box Plot")

plt.show()

print("Points outside the whiskers are outliers.")
```

Output

```
PS D:\FDS>  d:; cd 'd:\FDS'; & 'C:\Program Files\Python312\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\de
y\launcher' '56103' '--' 'D:\FDS\Hands-On_DS4.py'
Points outside the whiskers are outliers.
PS D:\FDS>
```



5. **Write a Python program to detect outliers in students' marks using the Interquartile Range (IQR) method.**

Dataset: student_marks.csv

```
Student_ID,Marks
S1,45
S2,50
S3,55
S4,58
S5,60
S6,62
S7,65
S8,95
```

Code

```
import pandas as pd

df = pd.read_csv("student_marks.csv")

Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Marks"] < lower) | (df["Marks"] > upper)]

print("Outliers")
print(outliers)
```

Output

```
PS D:\FDS>  d:; cd 'd:\FDS'; & 'C:\Program Files\Python312\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64
y\launcher' '56122' '--' 'D:\FDS\Hands-On_DS5.py'
Outliers
  Student_ID  Marks
7         S8     95
PS D:\FDS>
```

6. **Write a Python program to remove outliers from a sales dataset using the IQR method and display the cleaned dataset.**

Dataset: sales_data.csv

```
Sale_ID,Sales
S1,1200
S2,1500
S3,1700
S4,1600
S5,1800
S6,2000
S7,9000
```

Code

```
import pandas as pd

df = pd.read_csv("sales_data.csv")

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

print("Cleaned Dataset")
print(cleaned)
```

Output

```
PS D:\FDS>  d:; cd 'd:\FDS'; & 'C:\Program Files\Python312\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugp
y\launcher' '63072' '--' 'D:\FDS\Hands-On_DS6.py'
Cleaned Dataset
  Sale_ID  Sales
0      S1   1200
1      S2   1500
2      S3   1700
3      S4   1600
4      S5   1800
5      S6   2000
```

7. **Write a Python program to plot histograms and box plots before and after removing outliers from a dataset.**

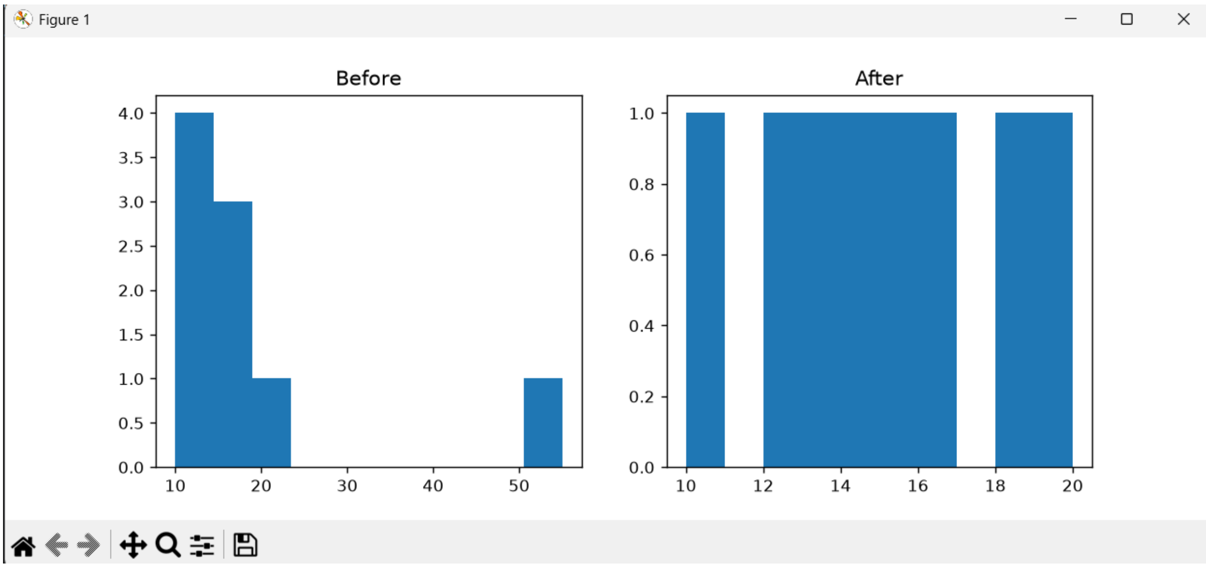
Dataset: data_values.csv

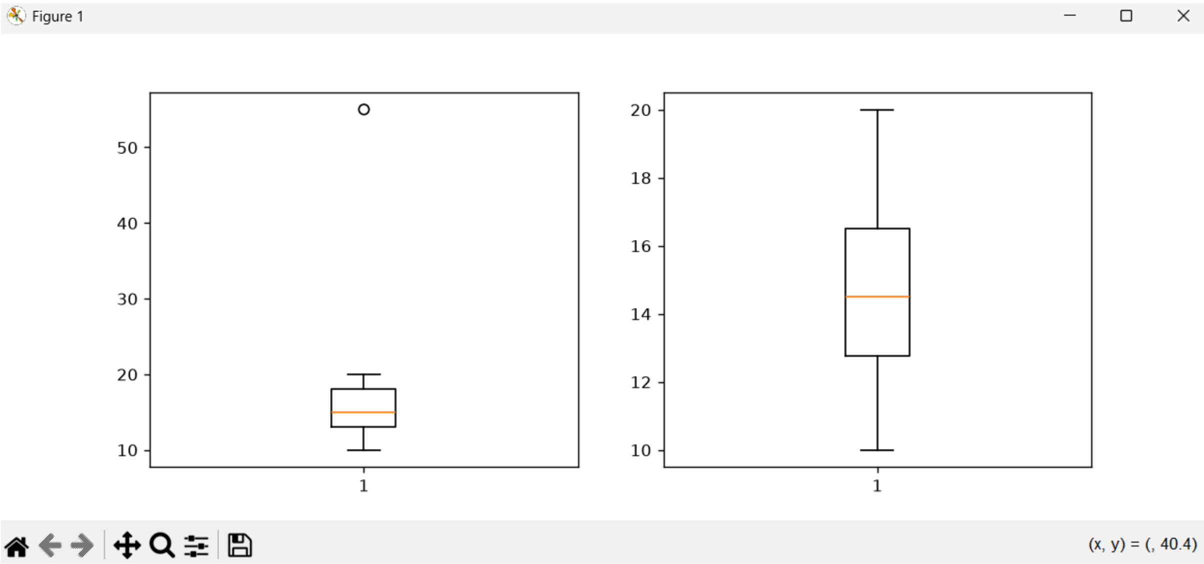
```
Value
10
12
13
14
15
16
18
20
55
```

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data_values.csv")

Q1 = df["Value"].quantile(0.25)
Q3 = df["Value"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned = df[(df["Value"] >= lower) & (df["Value"] <= upper)]

plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
plt.hist(df["Value"])
plt.title("Before")

plt.subplot(1,2,2)
plt.hist(cleaned["Value"])
plt.title("After")

plt.show()

plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
plt.boxplot(df["Value"])

plt.subplot(1,2,2)
plt.boxplot(cleaned["Value"])

plt.show()
```

Output





8. **Download the Iris or Titanic dataset and perform Exploratory Data Analysis by:**

- **Displaying dataset information**
- **Finding missing values**
- **Computing descriptive statistics**
- **Plotting histograms**
- **Plotting box plots**
- **Detecting outliers**
- **Removing outliers**
- **Saving the cleaned dataset**

Dataset

Iris.csv from kaggle

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("iris.csv")
print(df.info())
print("\nMissing Values")
print(df.isnull().sum())
print("\nDescriptive Statistics")
print(df.describe())
df.hist(figsize=(8,6))
plt.show()
df.boxplot(figsize=(8,6))
plt.show()
numeric = df.select_dtypes(include="number")
Q1 = numeric.quantile(0.25)
Q3 = numeric.quantile(0.75)
IQR = Q3 - Q1
cleaned = numeric[
    ~((numeric < (Q1 - 1.5 * IQR)) |
      (numeric > (Q3 + 1.5 * IQR))).any(axis=1)
]
cleaned.to_csv("iris_cleaned.csv", index=False)
print("Cleaned dataset saved as iris_cleaned.csv")
```

Output

```
PS D:\FDS>  d:; cd 'd:\FDS'; & 'C:\Program Files\Python312\python.exe' 'c:\Users\ACER\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debug
y\launcher' '58842' '--' 'D:\FDS\Hands-On_DS8.py'
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    str
dtypes: float64(4), str(1)
memory usage: 7.2 KB
None

Missing Values
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64

Descriptive Statistics
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.054000      3.758667     1.198667
std        0.828066     0.433594      1.764420     0.763161
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
Cleaned dataset saved as iris_cleaned.csv
```

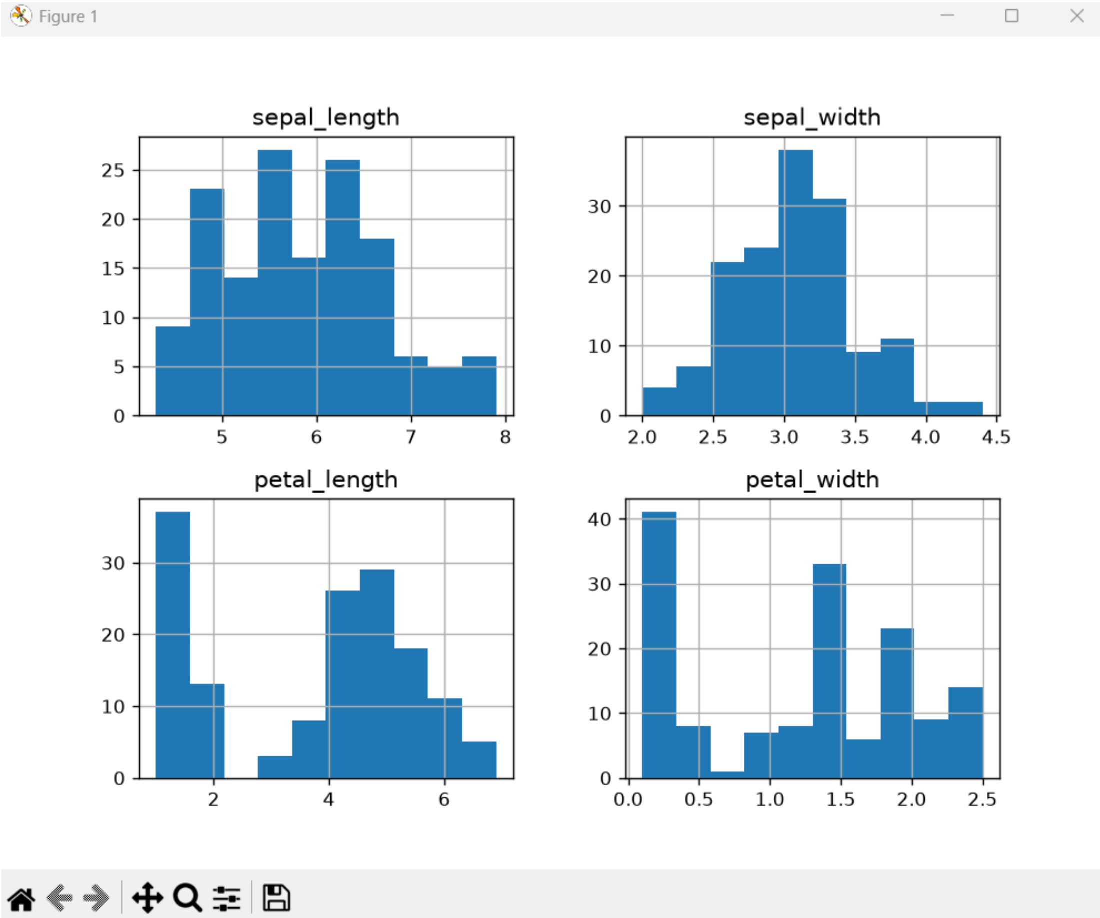


Figure 1
sepal_length
sepal_width
petal_length
petal_width

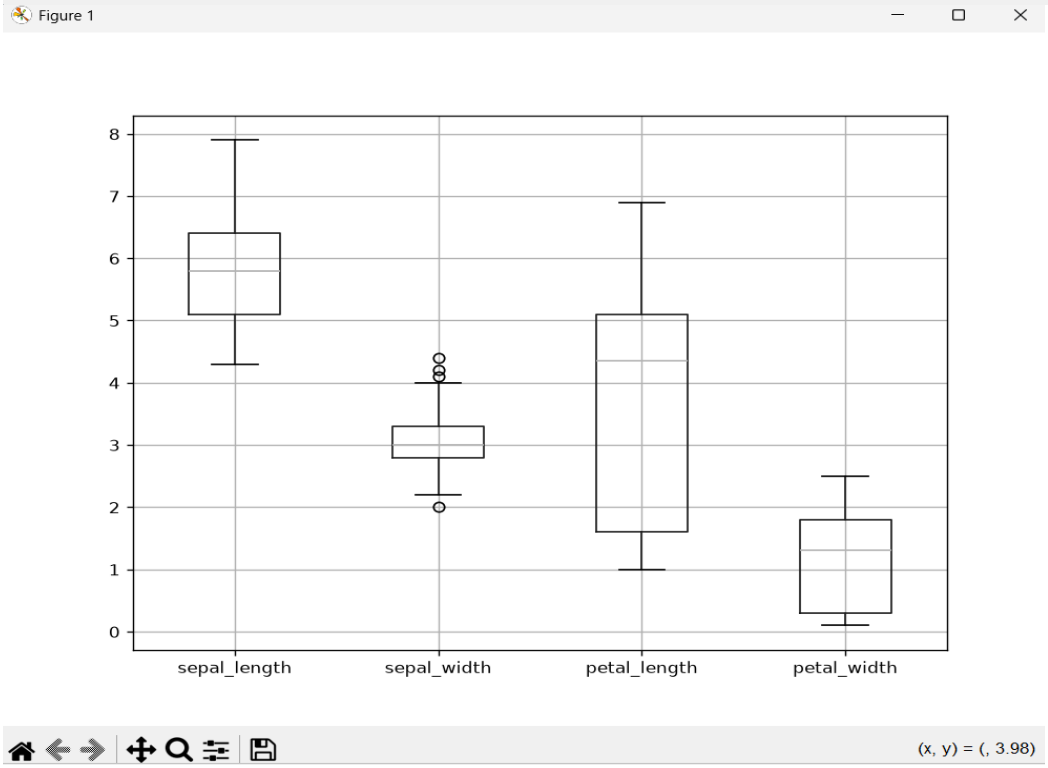


Figure 1
sepal_length
sepal_width
petal_length
petal_width
(x, y) = (, 3.98)